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31 December 1985
SUPERSEDING
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7 February 1977

MILITARY SPECIFICATION

TEST SET, LOW VOLTAGE CIRCUIT, AUTOMOTIVE, 12 AND 24 VOLT DC SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of low voltage circuit test set for testing standard electrical systems incorporated in commercial and military automotive vehicles, including tanks (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

W-C-440	-	Clips, Electrical, General Specification for.
DD-G-1403	-	Glass, Float, Sheet, Figured, Coated (Heat-Strengthened and Tempered).

MILITARY

MIL-W-80	-	Window, Observation, Acrylic Base, Antielectrostatic, Transparent (For Indicating Instrument).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-T-152 - Treatment, Moisture- and Fungus-Resistant. Of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment.
MIL-G-3787 - Glass, Laminated, Flat; (Except Aircraft).
MIL-C-13486 - Cable, Special Purpose, Electrical: Low-Tension Heavy Duty Single-Conductor and Multiconductor.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-889 - Dissimilar Metals.

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380-84 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The test set shall be new and of the manufacturer's latest design meeting the performance requirements specified herein. The set shall be designed to test low-voltage circuits on 12 and 24 volt direct current (DC) standard electrical systems. Those parts of the test set subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the test set or any set component. Asbestos or products containing asbestos shall not be used in the test set.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the test set. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380-84.

3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.2.3 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.2.4 Safety and health requirements. The test set shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the test set itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other

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used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4 Construction. The test set shall be furnished complete so it can be used for the operations specified herein. Parts used in fabricating the test set shall be new and free of defects. All part surfaces shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting performance.

3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed. Screw threads mating in aluminum shall be given an application of anti-seizing compound.

3.4.2 Electrical connections. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. All other electrical connections shall be mechanically secured and soldered or brazed. Only rosin base fluxes shall be used in soldering operations. Soldering shall be in accordance with Requirement 5 of MIL-STD-454.

3.5 Performance.

3.5.1 Accuracy of internal circuits and test leads. Accuracy of internal circuits and test leads shall be determined in an ambient air temperature of +80° Fahrenheit (F) ±5°F with the test set placed in a horizontal position.

3.5.1.1 Initial accuracy. With the test set horizontally positioned in an ambient temperature of +80° ±5°F, the over-all initial accuracy shall be within seven percent of the full scale value for the 500 amp shunt ammeter circuit and three percent of the full scale value for the internal shunt ammeter circuits. Voltmeter system accuracy shall be ±two percent of full scale reading on all ranges.

3.5.1.2 Reliability. The over-all reliability in accuracy shall be within one percent of the full scale value of the initial accuracy readings for both the ammeter and voltmeter circuits.

3.5.2 Temperature and humidity conditions.

3.5.2.1 Operating temperatures. The meters shall indicate freely throughout the full range of the scale when the test set is operated in ambient air temperatures of +30° ±3°F to +112° ±3°F. The test set shall conform to the accuracy requirement of 3.5.1.2 while operating at those temperatures.

3.5.2.2 Humidity conditions. The test set shall operate within the accuracy requirements specified in 3.5.1.2 after being subjected for 20 hours to a relative humidity of 93 to 97 percent in an air temperature of +75° ±2°F, air movement and radiation negligible, followed by four hours exposure to a relative humidity of 15 to 20 percent in an air temperature of +112° ±3°F, air movement and radiation negligible.

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3.5.3 Momentary overload. Each meter and its associated circuit shall be sufficiently rugged to withstand, without damage, a series of momentary overloads. It shall not be necessary to make any repairs or adjustments other than resetting the pointer to zero through the external zero adjuster.

3.5.3.1 Ammeter circuit. The ammeter and its circuit, with meter protection device, shall withstand a momentary overload of five times end-scale value for one second ON and 60 seconds OFF, for a total of five cycles, followed by a five second ON period.

3.5.3.2 Voltmeter circuit. The voltmeter and its associated circuit, with meter protection device, shall withstand a momentary overload of 2.8 times full-scale value for the periods specified in 3.5.3.1.

3.5.4 Sustained overloads. The meters and their associated circuits, with meter protection devices, shall withstand an overload 20 percent greater than end-scale value for a continuous period of eight hours without damage or a change in overall reliability greater than specified in 3.5.1.2.

3.5.5 Vibration. The test set shall withstand a vibration of simple harmonic motion having an amplitude of 1/32 inch (1/16 inch maximum total excursion) at a frequency range of 5 to 55 cycles per second (cps). The frequency range shall be traversed once per minute. The test set shall withstand this vibration for one hour in each of the three major axes. Screws, bearings, pivots, windows, and other parts shall not be loosened or damaged. Accuracy shall comply with 3.5.1.2.

3.5.6 Shock. The test set shall withstand the shock resulting from being dropped a vertical distance of eight inches on a concrete surface in each of its major axes. Screws, bearings, pivots, windows, and other parts shall not be loosened, cracked, or damaged. Accuracy shall comply with 3.5.1.2.

3.5.7 Dielectric strength. The test set shall withstand a potential of 500 volts root mean square (rms), 60 hertz (Hz) for one minute without showing evidence of smoking, arcing, flashover, charred insulation, or other damage caused by high voltage application. The specified potential voltage shall be applied between conductor to ground, conductor to conductor, mutually insulated portions of the test set, or insulated portions and ground. All controls and switches shall be in the OFF position during potential application.

3.5.8 Case leakage. The case (see 3.6.1), with the cover closed, shall show no evidence of leakage at, or above, the control/meter panel elevation when subjected to a spray of water at a rate of 5 $\pm$1/2 gallons per hour (gph). The spray shall be spread uniformly over the top of the case at a 45 degree angle from verticle.

3.6 Components. The test set shall consist of the principal components specified in 3.6.1 through 3.6.7.

3.6.1 Case. The case shall include compartments, an instrument panel, carrying handles, rubber (machine) bumpers, hinged removable cover, and tension cover fasteners. The maximum overall dimensions of the case shall be 18 inches wide across the front, 15 inches long front to back, and 12 inches high. The overall dimensions shall include the rubber (machine) bumpers, hinges, and carrying handles. The case shall be provided with louvers or expanded metal

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screening for dissipation of heat from the load bank. Drain holes, 1/4 inch in diameter, shall be placed in the bottom of each corner to vent and drain the space below the panel. The case shall be constructed of steel or aluminum not less than Manufacturers' Standard Gage No. 20 (0.036 inch), except as specified in 3.6.1.1 and 3.6.1.2. Carrying handles shall be provided on each end of the case and shall be of sufficient size to provide a comfortable hand grip. If wire type handles are furnished, they shall be 1/4 inch diameter minimum at the grip part of the handle. A complete circuit diagram, including component values and part numbers, shall be legibly marked on the inside bottom of the case in such a place that it is not visible until the instrument panel is removed. Rubber (machine) bumpers shall be provided on the outside bottom near each corner.

3.6.1.1 Compartments. The test set shall be compartmentalized for test lead storage and for the load unit and control/meter panel. The test lead compartment shall be designed for storage and ease of removal of all test leads and the external shunt. Partitions between all compartments shall be constructed of steel or aluminum not less than Manufacturers' Standard Gage No. 21 (0.033 inch).

3.6.1.2 Instrument panel. The instrument panel shall be constructed of steel or aluminum not less than Manufacturers' Standard Gage No. 20 (0.036 inch). The field locations for the function items of the control/meter panel shall be within the confines shown in Figure 1. (Figure 3 is provided only as a reference to depict a desired control/meter layout, with emphasis on the eleven terminals within their field area.) All electrical component parts shall be mounted directly on the panel, or on a chassis attached to the panel, so that when the panel is removed from the case all components are removed from the case. All binding posts and switches passing through the instrument panel shall be insulated from the panel to withstand the mechanical and electrical stresses specified herein. Dials, markings, and controls on the instrument panel shall be in the normal reading position when the observer faces the 18 inch side of the test set and the top of the test set, which is the area bounded by the 18 by 15 inch dimension (test set being 12 inches high). All markings on the instrument panel shall be legibly and permanently marked in accordance with MIL-STD-130. The panel marking for the field rheostat control shall read INCREASE GENERATOR OUTPUT with an arrow pointing in the direction the knob is to be rotated to decrease resistance. The rheostat panel shall also be marked DO NOT EXCEED 15 AMPS. The open position for the rheostat shall be marked on the panel. The panel markings for the load bank switch shall read OFF and ON. The panel markings for the load bank controls shall read INCREASE LOAD with an arrow pointing in the clockwise direction to decrease resistance. The panel markings for the binding posts shall be as shown in Figure 2. The panel markings for the load bank and field rheostat shall include the following:

DUTY CYCLE: One hundred (100) percent.

CAUTION: Always turn the load bank switch to OFF position before attaching or removing load bank test leads.

3.6.1.3 Removable cover. The removable cover shall be hinged to the case on one end and fastened to the case by two tension fasteners. One on each of the adjacent sides or two on the end opposite the hinges. A storage compartment for an instruction handbook shall be located in the cover.

3.6.2 Ammeter and voltmeter. One ammeter and one voltmeter shall be flush mounted on the instrument panel. The flush mounted meters shall have their major

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portions extended below the mounting panel surface. A shock resistant seal shall be provided between the instrument panel and the meter housing. Exposed surfaces of meter housings shall have a dull or matte finish.

3.6.2.1 Meter pointer. The meters shall have a pointer formed to permit accurate readings from a normal 12 inch distance, and approximate readings from a distance where dial scale markings are legible. The pointer shall be of rigid construction. The pointer shall withstand all of the tests specified herein without becoming damaged or distorted. When the test set is not provided with a means for avoiding errors in reading due to parallax, the clearance between the pointer index and the graduated scale shall not exceed 0.10 inch or be less than 0.03 inch.

3.6.2.2 Meter protection.

3.6.2.2.1 Meter movement protection. The meters shall be protected against damage due to error in lead hookup or from any malfunction which may occur within the test set circuitry. The meter protection shall be equivalent to "Lectrotech Metergard" or "Simpson Meter Safe/Guard Model No. 12621", or equal, and shall be installed internally within the ammeter. The protective device may be installed either internally or externally to the voltmeter. The device shall not affect the accuracy of the meters and circuits in their normal operating ranges.

3.6.2.3 Windows. Unless otherwise specified (see 6.2.1), the test set shall be provided with windows of glass or plastic material, and shall be free from discoloration, scratches, striae, electrostatic effect, or other detrimental defects which may interfere in any way with normal reading of the dial scales. Glass windows, when used, shall comply with MIL-G-3787, Class II, or DD-G-1403, Kind Ft or HS, Condition A. Plastic windows, when used, shall be flat sheet or preformed acrylic meeting the requirements of MIL-W-80. The plastic windows shall be treated on both sides with an anti-static solution, silicon base preferred. The windows shall be secured with clamps or other suitable devices. Use of cement alone for holding the windows in place shall not be acceptable.

3.6.2.4 Scale legibility. The scale shall be legible at a distance of 18 inches.

3.6.2.5 Adjustment. Meters shall be provided with an external zero adjustment. The range of the zero adjustment, both above and below the center of the zero mark, shall be not less than three percent of the end scale length. Both the meter and zero adjustment shall not be loosened or damaged when subject to vibration (see 3.5.5).

3.6.2.6 Dial. The meter dials shall be made of nonferrous metal. The face of the dial shall have a bonded, dull white background with the numerals and lettering printed thereon. The mean scale arc length shall be not less than 3.2 inches, with the full scale covering an arc not less than 85 degrees nor more than 105 degrees.

3.6.2.6.1 Ammeter dial. The ammeter dial shall have one scale and shall indicate the following scale ranges: -3 to 0 to +10, -15 to 0 to +50, -30 to 0 to +100, and -150 to 0 to +500. The scale shall be divided into 130 divisions, with heavy extended lines for each five divisions and numerals for each ten divisions. Maximum end scale deflection shall be 50 millivolts (mv) and shall be noted on the face of the dial as FS-15-0-50-MV.

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3.6.2.6.2 Voltmeter dial. The voltmeter dial shall have one scale and shall indicate the following scale ranges: 0 to +1, 0 to +10, 0 to +20, and 0 to +50. The scale shall be divided into 50 divisions with heavy extended lines and numerals for each five divisions. Full scale (FS) deflection shall be one milliampere (MA), and shall be noted on the face of the dial as FS=1 MA.

3.6.2.7 Position influence (balance). When the meters are tilted 60 degrees from the horizontal position in any direction, the pointer of either meter shall not deviate from the zero mark more than one percent of the end scale length.

3.6.2.8 Damping factor. The damping factor shall be not less than 2.5. The damping factor shall be the ratio of the steady deflection, in angular degrees, to the difference between the maximum momentary deflection and the steady deflection in angular degrees.

3.6.2.9 Response time. The length of time required for the pointer of each meter to come to apparent rest, when steady electric power sufficient to produce momentary end scale deflection is applied, shall not exceed two seconds.

3.6.2.10 Meter power consumption. The power consumption required to produce end scale deflection shall be 50 mv $\pm$2 percent for the ammeter, and one ma $\pm$1 percent for the voltmeter.

3.6.3 Binding posts. Eleven non-ferrous metal binding posts shall be provided. The finding posts shall be cross-drilled and reamed to receive the jack plugs and shall include a means to prevent the screws from being removed from the posts. The posts shall be mounted on the control/meter panel within the field locations shown in Figure 2. The binding posts shall secure the jacks in a positive manner and shall be capable of carrying, continuously, the maximum current of the circuit involved.

3.6.4 Ammeter circuit. The ammeter circuit shall have shunts, an ammeter, variable load bank (with the blower motor to be a part of the total load), load disconnect switch, binding posts, and necessary switches or shorting bars to select proper ammeter ranges. The ammeter circuit shall allow use of the ammeter and the load bank, independent of each other and in series with each other.

3.6.4.1 Ammeter shunts. Shunts for all ammeter ranges, except the 500 ampere range, shall be internal. The external shunt for the 500 ampere scale range shall conform to the requirements specified herein, shall withstand 500 amperes applied for ten minutes, and shall maintain an accuracy within one percent of maximum shunt capacity. One pair of calibrated shunt leads, 108 inches long, shall be furnished and permanently attached to the shunt on one end. The other end shall terminate in solid jack plugs of suitable size to fit their respective binding posts. The calibrated leads shall extend from the shunt in the same direction. High current carrying leads, 12 inches in length, shall be provided and permanently attached to the shunt on one end. The other end shall have spring-type clips with insulators for making test connections. The high current leads, clips, and insulators shall be capable of carrying rated current for a period of ten minutes without damage to insulation, wire clips, or insulators. The shunt shall be protected by a wrapping of suitable electrical covering.

3.6.4.2 Resistive load unit. A variable resistive load unit, consisting of several fixed resistors and a variable resistor (rheostats) or consisting of two

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or more variable resistors (rheostats), shall be provided and shall be so designed that any of the following loads may be obtained at the voltages specified over a current range of five to 100 amperes within one ampere increments in both ascending and descending order:

AMPERES
(within 0.50 ampere increments)

Volts	Amperes	Watts
15.0	5 to 100	75 to 1500
30.0	5 to 100	150 to 3000

Excluding the load unit disconnect switch, a combined total of not more than four switches or strap links shall be used to control and vary the load. When the resistive load unit consists of two or more variable resistors (rheostats), variable resistance of equal ohmic value and wattage rating connected in series, or parallel, shall be coupled so that when the resistance is varied each resistor shall absorb an equal share of the load within ±15 percent. The wattage rating of the variable resistances shall be large enough to prevent overheating if the load becomes unbalanced within the limits specified. The load unit shall be connected to terminal binding posts insulated from the control panel. Provisions shall be made to prevent lead connection to any strapped binding post. Strap links, if used, shall be permanently secured on one binding post and the strap area of contact shall be of sufficient size to carry, continuously, the maximum current of the circuit involved. When the load unit is connected for operation and adjusted for a load current of ten to 100 amperes, the load unit shall be capable of maintaining each set value of load within ±15 percent during ten minutes of continuous operations without readjustment of the load controls. Operating the test set at 30 volts and 100 amperes for one full hour shall not cause a temperature rise of more than 40°F above ambient temperature at any point on the interior wall of the lead compartment or any exterior surface of the test set. The load unit shall withstand without damage a current of 100 amperes at 30 volts for a period of eight continuous hours without interruption.

3.6.4.3 Load bank switch. A load bank switch shall be provided to disconnect the load. The switch shall withstand 10,000 cycles of operation at the rate of ten cycles per minute (cpm) while carrying 100 amperes at 30 volts DC.

3.6.4.4 Variable resistor endurance. All variable resistors (rheostats) shall operate satisfactorily after 1,000 cycles of operation at a rate of 30 cycles per hour under maximum rated current. The variable resistors, during each two minute cycle, shall be operated from the OFF position to the maximum current position and back to the OFF position within 30 seconds, followed by a 90 second OFF time.

3.6.4.5 Cooling fan and connections. The test set shall be provided with an internal cooling fan powered by a direct current motor capable of operating from 15 and 30 volts DC. The power connections shall be internal and shall cause the fan motor to activate when a load is applied.

3.6.5 Voltmeter circuit. The voltmeter circuit shall have individual, fixed, precision resistors for each voltage range. The design shall insure the continued use of the voltmeter on other ranges if any one resistor should open the circuit in operation.

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3.6.5.1 Scale selection. When a voltmeter selector switch is furnished, it shall be a five position switch reading from left to right: OFF-50-20-10-1. Binding posts shall be as specified in 3.6.3. When an ammeter selector switch is furnished, it shall be a four position switch reading from left to right 100A-50A-10A-EXT. All switch contacts shall be silver plated.

3.6.6 Field rheostat circuit. A built-in, knob controlled, variable (within 0.20 ampere) rheostat of 0 to 70 ohms, 0.2 to 15 amperes, taper wound, with a continuous duty rating shall be provided and shall be connected to two binding posts. The rheostat shall contain an open position at the high resistance end of the rheostat. The rheostat shall be capable of carrying ten amperes at 0.5 ohms when 30 volts are applied, with field of 2.5 ohms resistance and rheostat in series.

3.6.7 Test leads. All test leads shall be single conductor cable conforming to type I requirements of MIL-C-13486. All leads shall be identified. Clips and insulators on leads listed below shall conform to type PC and PCL requirements of W-C-440. For items (a), (b), and (c) below, the connector insulators and clip insulators shall be red for positive (+) leads and black for negative (-) leads:

(a) 2 - Voltmeter leads, AWG No. 16 cable minimum size, 108 inches long, with clips and insulators on one end and solid jack plugs of a size to fit their respective binding posts on the other end.
(b) 2 - Ammeter leads, AWG No. 6 cable minimum size, 108 inches long, with clips and insulators on one end and solid jack plugs of a size to fit their respective binding posts on the other end.
(c) 2 - Field rheostat leads, AWG No. 16 cable minimum size, 108 inches long, with clips and insulators on one end and solid jack plugs of a size to fit the respective binding posts on the other end.
(d) 1 - Jumper lead, AWG No. 16 cable minimum size, 12 inches long, with clips and insulators on both ends.

3.7 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.8 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry.

3.9 Product identification. The test set shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.10 Workmanship. Standards of workmanship shall assure that the test set shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of

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all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Accuracy tests (see 4.6.1.2 and 4.6.1.2.1).
(c) Load bank resolution test (see 4.6.5).
(d) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105, using the following Inspection Levels and Acceptable Quality Levels (AQL):

Inspection or Test	Inspection Level	AQL
Examination (see 4.5)	100 percent	-
Accuracy tests (see 4.6.1.2 and 4.6.1.2.1)	II	1.0
Load bank resolution test (see 4.6.5)	II	1.0
Packaging inspection (see 4.7)	S-4	4.0

NOTE: Lot size shall be no more than 150 test sets.

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4.5 Examination. Visually and manually examine the test set to determine conformance with the requirements of 3.2 through 3.4.2 and 3.6 through 3.10. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of moveable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the test set and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Performance tests. All performance tests shall be conducted in an ambient air temperature of +80° ±5°F.

4.6.1.1 Meter movement protection tests. The ammeter and voltmeter shall be tested as specified in 4.6.1.1.1 and 4.6.1.1.2 to determine compliance with 3.6.2.2.1. Following those tests, meter accuracy tests shall be conducted as specified in 4.6.1.2 and 4.6.1.2.1.

4.6.1.1.1 Ammeter protection test. The ammeter shall be disconnected from its associated test set circuitry and connected to a 0-3 volt DC voltmeter having a sensitivity of 100 ohms per volt or greater across the terminals. A 250 ma DC current shall be applied momentarily to the ammeter terminals. The voltage shall not exceed 1.5 volts. The polarity shall be reversed and the test repeated.

4.6.1.1.2 Voltmeter protection test. With the voltmeter on the one volt range, 30 volts DC shall be applied momentarily to the test leads. The current through the meter and its associated circuitry shall measure not less than 30 ma to verify shunting capability of the meter protection device. The polarity shall be reversed and the test repeated.

4.6.1.2 Accuracy of ammeter and voltmeter circuits and test leads. The accuracy tests shall be conducted with the test set placed in a horizontal position. All current measurements taken throughout the range of five through 100 amperes shall be measured with the current flowing through the test set load bank circuit. The range of 100 through 500 amperes shall be measured with the current flowing through an external load. The test set ammeter shall be used to measure the current during all tests.

4.6.1.2.1 Initial accuracy test. Prior to any testing, the initial accuracy of the ammeter and voltmeter with their internal circuits, including test leads, shall be determined. The accuracy of the circuits shall be measured on each scale range. The test readings shall cover a minimum of six points equally spaced over the entire scale of each scale range to determine compliance with 3.5.1.1. Tapping of meters shall be permitted only for the zero settings. Accuracy measurements shall be compared with meters that are accurate within 0.25 of one percent.

4.6.1.2.2 Reliability test. After completion of all the tests for the requirements specified in 3.5.2 through 3.5.7 and with the test set stabilized at the ambient air temperature specified in 4.6.1, the reliability accuracy shall be

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determined. The accuracy shall be measured in accordance with the testing procedure specified in 4.6.1.2.1 to determine compliance with 3.5.1.2.

4.6.1.3 Damping factor. Electric power sufficient to produce momentary full scale deflection shall be applied to the meters. The maximum momentary deflection shall be observed and recorded. After the pointer has come to rest, the steady deflection shall be observed and recorded to determine compliance with 3.6.2.8. The damping factor shall be taken as the ratio of the steady deflection, in angular degrees, to the difference between the maximum momentary deflection and the steady deflection in angular degrees.

4.6.1.4 Response time. Steady electric power sufficient to produce momentary full scale deflection shall be applied to the meters. The length of time, in seconds, required for the pointer to come to apparent rest shall be recorded to determine compliance with 3.6.2.9. The pointer shall be considered as having come to apparent rest when it has attained the actual point of rest within one percent of the full scale value. The test shall be repeated and recorded at least four times for each meter. The average of the recorded readings of a meter shall be considered as the response time for that meter.

4.6.1.5 Position influence. The pointer shall be set to the center of the zero scale division with the meter in the normal operating position. The meter shall then be tilted 60 degrees from this position in any direction and the deviation of the pointer shall be noted to determine compliance with 3.6.2.7.

4.6.1.6 Zero adjustment. The range of the zero adjustment above and below the center of the zero mark shall be observed to determine compliance with 3.6.2.5. Record shall be made as to whether or not the zero adjuster can be rotated through 360 degrees without damage to the meter.

4.6.1.7 Meter sensitivity. The voltage required to produce full scale deflection of the ammeter and the current required to produce full scale deflection of the voltmeter shall be measured at the meter terminals. Failure of the measurements to meet the requirements of paragraph 3.6.2.10 shall be cause for rejection.

4.6.1.8 Dielectric strength. The test set shall be subjected to 500 rms volts, 60 Hz between all external conductors and the following:

(a) Meter panel.
(b) Zero adjuster.
(c) Switches and load controls.

Voltage shall be applied at the rate of approximately 250 volts per second until the specified voltage is reached. Voltage shall be maintained at 500 volts for a period of 60 seconds. The test set shall then be inspected for conformance with 3.5.7.

4.6.2 Humidity test. The test set shall be subjected to 20 hours exposure at a relative humidity of 93 to 97 percent in an air temperature of +75° ±2°F, air movement and radiation negligible (see 3.5.2.2). While at this condition, the test set shall be tested to determine compliance with the accuracy requirements specified in 3.5.1.2. Following the above, the test set shall be subjected to four hours exposure while at a relative humidity of 15 to 20 percent in an air temperature of +112° ±3°F, air movement and radiation negligible. While at this

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condition, the test set shall be tested to determine compliance with the accuracy requirements specified in 3.5.1.2.

4.6.3 Operating temperature tests.

4.6.3.1 Low ambient air temperature test. After being in an ambient air temperature of +30° ±3°F until stabilized, and while at that temperature, the test set shall be tested for accuracy in accordance with the testing procedure specified in 4.6.1.2.1 to determine compliance with 3.5.2.1.

4.6.3.2 High ambient air temperature test. After being in an ambient air temperature of +112° ±3°F until stabilized, and while at that temperature, the test set shall be tested for accuracy in accordance with the testing procedure specified in 4.6.1.2.1 to determine compliance with 3.5.2.1.

4.6.4 Case leakage test. The test set, with the cover closed and while resting on the bottom rubber (machine) bumpers, shall be tested to determine compliance with 3.5.8. The case, with the panel inside, shall be subjected to rain spray nozzles of such size, and so spaced that 5 ±1/2 gph of water falls for one hour, uniformly distributed, over the top of the test set case.

4.6.5 Load bank resolution test. Under an ambient air temperature of +80° ±5°F, the load bank shall be tested to determine compliance with the resolution requirements of 3.6.4.2 as follows:

(a) The load bank shall be connected to a 15 volt power source. The resistance of the load shall be adjusted to increase the current from 5 to 100 amperes in increments no greater than 1.0 ampere. The current reading shall be recorded after each incremental change.
(b) After cooling for ten minutes, the load bank shall be connected to a 30 volt power source and the test repeated as specified in (a) above.
(c) After cooling for ten minutes, the field rheostat shall be adjusted to 0.5 ohm. Ten amperes shall be passed through this resistance for ten minutes. The load bank shall show no damage due to arcing, flashover, or excessive heating at the conclusion of these tests (see 3.6.6).

4.6.6 Load division test. If two or more variable load resistors (rheostats) of equal value are connected in series, a voltage measurement shall be taken across each resistor, at the various voltage inputs, when adjusted to 100 amperes. If the variable load resistors (rheostats) are connected in parallel, a current measurement shall be taken through each resistor, at the various voltage inputs, when adjusted to a total current of 100 amperes. The voltage drop or current, as applicable, to each variable resistor (rheostat) shall be such that the load absorbed by each resistor shall be the same within ±15 percent for a ten minute period (see 3.6.4.2).

4.6.7 Momentary overload. The test set shall be subjected to the following momentary overload tests (see 3.5.3.1 and 3.5.3.2):

(a) Ammeter circuit: the 3,000-watt resistance load bank shall be used in conjunction with the -3 to 0 to +10 ammeter range. The load bank shall be set to absorb a current of 50 amperes at 28 volt input.

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This input shall be applied to the circuit for one second ON and 60 seconds OFF for a total of five cycles, followed by a five second ON period.

(b) Voltmeter circuit: concurrently with (a) above, the 0 to 10 voltmeter range shall be connected across the 3,000-watt resistance load bank. The applied momentary overload voltage shall be 28 volts direct current.

(c) Upon conclusion of the above tests, the accuracy of the -3 to 0 to +10 ammeter circuit and 0 to 10 voltmeter circuit shall be examined for damage and tested as in 4.6.1.2.2 to determine compliance with 3.5.3, 3.5.3.1, and 3.5.3.2.

4.6.8 Sustained overload. The test set, with its circuits interconnected as specified in 4.6.7 and with the exceptions specified herein, shall be subjected to an eight hour continuous period test. The load bank shall be set to absorb 12 amperes at 12 volts input. Upon conclusion of test, the overall accuracy of all ammeter and voltmeter ranges shall be tested in accordance with 4.6.1.2.2 to determine compliance with 3.5.4.

4.6.9 Vibration test. The test set shall be rigidly mounted on a vibration machine and shall be subjected to vibrations (simple harmonic motion) with an amplitude of 1/32 inch (1/16 inch total excursion) through a frequency range of 5 to 55 cps executed once per minute. The test set shall be vibrated for one hour in the direction of each of its three major axes. Upon conclusion of the test, the test set shall be examined for damage and tested in accordance with 4.6.1.2.2 to determine compliance with 3.5.5.

4.6.10 Shock. The test set shall be dropped a distance of eight inches on a concrete surface, once on each of its major axes. The test set shall be examined for damage and the overall accuracy for all meter ranges shall be checked in accordance with 4.6.1.2.2 to determine compliance with 3.5.6.

4.6.11 High current test. Five hundred amperes shall be passed through the 500 ampere shunt for a period of ten minutes. Accuracy shall be tested during the first and last minute of this test to determine compliance with 3.6.4.1.

4.6.12 Endurance and temperature rise test. Thirty volts shall be applied to the internal load bank and ammeter. The load bank shall be adjusted for a current reading of 100 amperes. This current shall be maintained for eight hours. The temperature of the exterior surfaces of the test set shall be noted at the beginning of the test and at the end of the first hour of operation. At no point on the test unit surface shall the temperature increase more than 40° Fahrenheit. Any visible effects of damage after the eight-hour test, such as burning due to arcing or flashover, cracked components or charred insulation shall be cause for rejection.

4.6.12.1 Load bank switch. The load bank switch shall be subjected to an endurance test of 10,000 cycles of operation at a rate of ten cpm to determine compliance with the requirements specified herein and in 3.6.4.3. Each test cycle shall consist of operating the switch from the OFF position to the ON position and then back to the OFF position. The switch shall satisfactorily make and break its rated current (100 amperes at 30 volts) throughout this test.

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4.6.12.2 Variable resistors. All variable resistors shall be subjected to an endurance test of 1,000 cycles of operation at a rate of 30 cycles per hour under maximum rated voltage to determine compliance with 3.6.4.4. Each two minute cycle shall consist of operating the variable resistance from the OFF position to the maximum current position and back to the OFF position within 30 seconds, followed by a 90 second OFF time.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation, Levels A or B packing, and military marking shall be in accordance with MIL-E-17555.

5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The test set covered by this specification is intended to be used for testing standard, low voltage, electrical systems (12 and 24 volts) in military and commercial vehicles. An adapter kit (NSN 4910-00-348-7600) is, however, required when testing waterproof electrical systems in military vehicles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.4).
(d) Test set windows if different (see 3.6.2.3).
(e) Painting and finishing if different (see 3.8).
(f) Product identification if different (see 3.9).
(g) First article inspection if different (see 4.3).
(h) Quality conformance inspection if different (see 4.4).
(i) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, should include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer), Federal Supply Code

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for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
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Air Force - 99

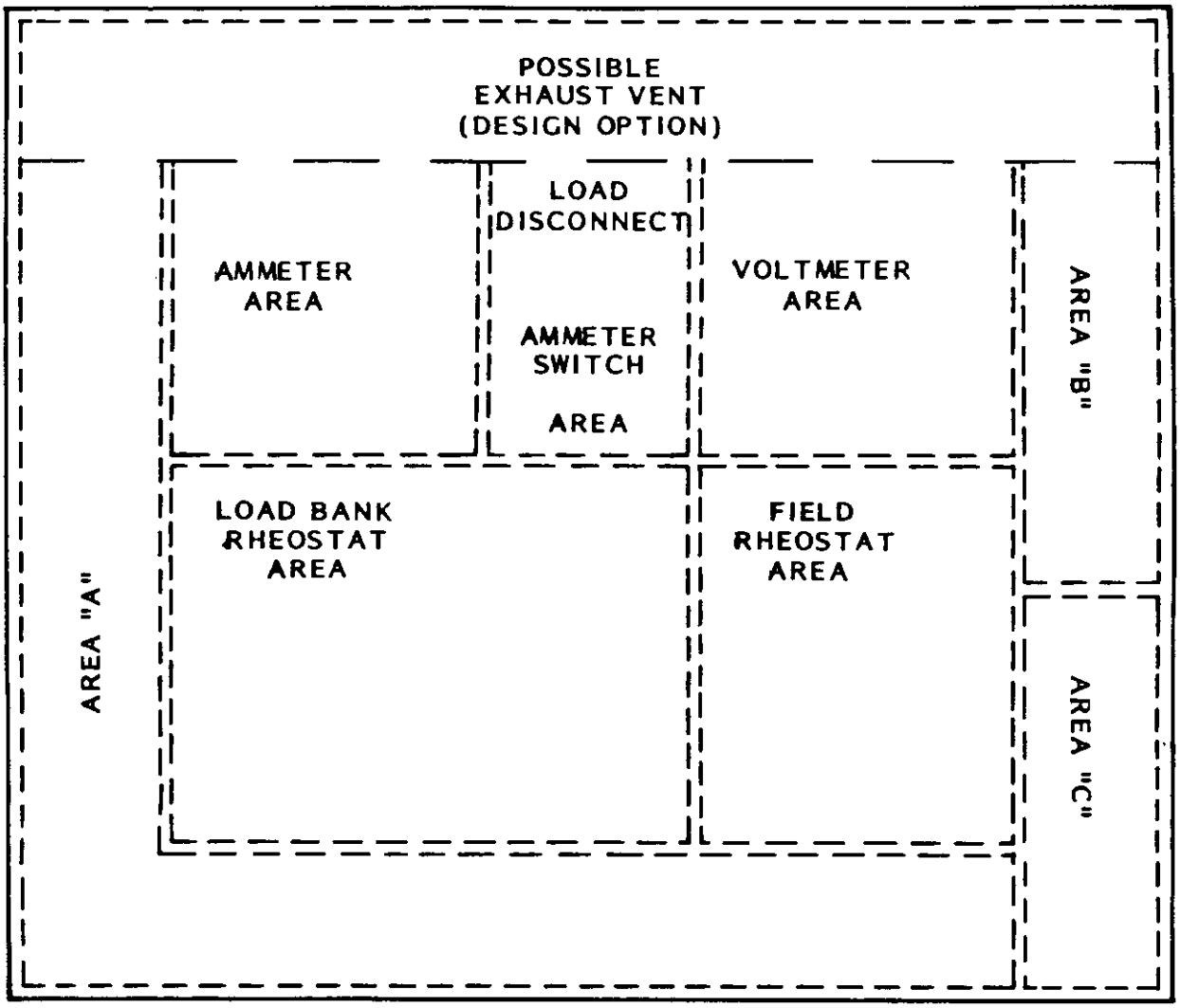
Preparing activity:
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Project No. 4910-0589

Review activity:
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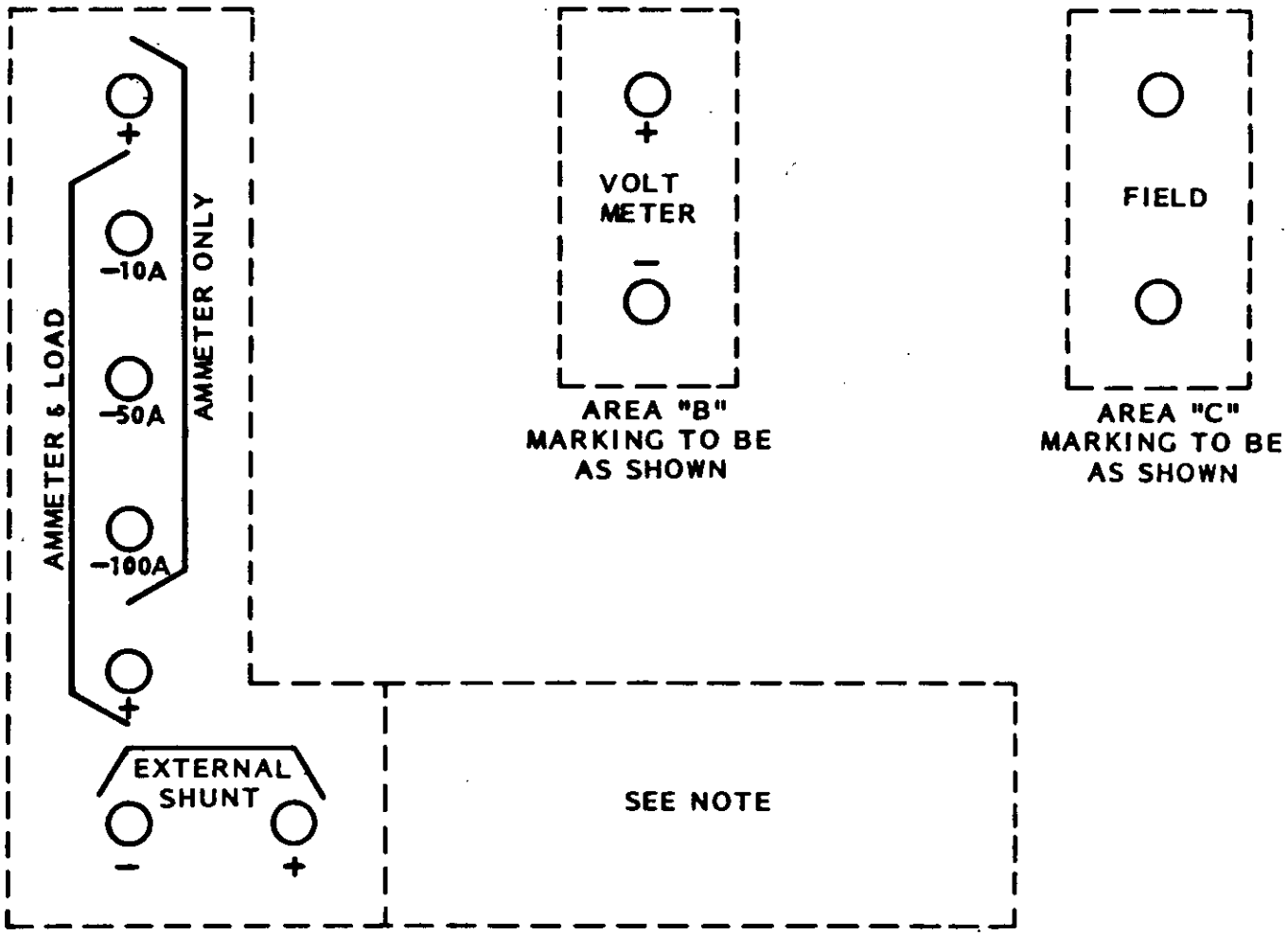
NOTES: 1. All function items shall be located within the confines of their respective (dotted line) fields.

2. Enclosed areas are relative, not dimensional.

3. Terminal lug identification marking shall be as shown in Figure 2.

FIGURE 1. Field locations.

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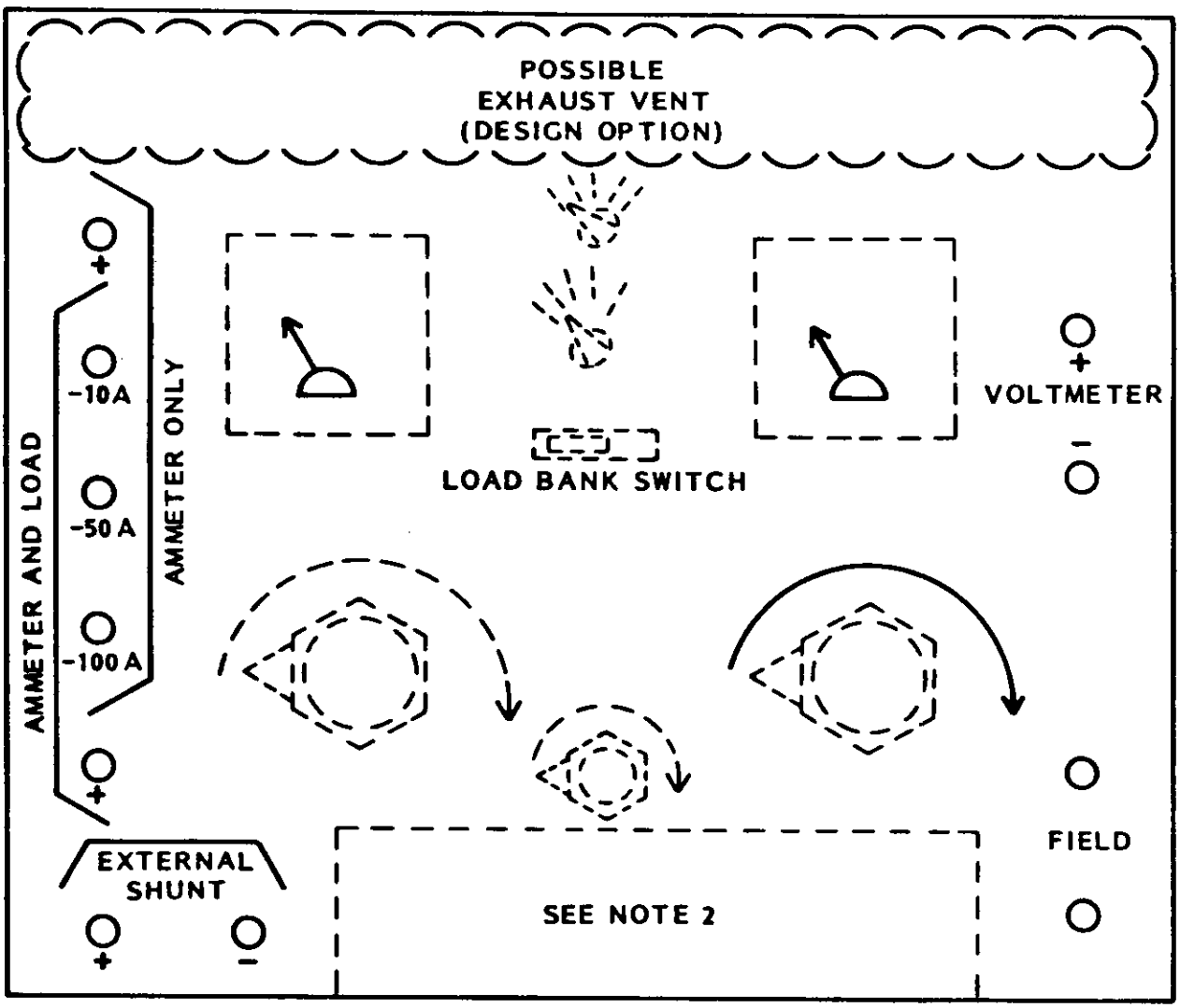


AREA "A" TERMINAL SHALL BE MARKED AS SHOWN

NOTE: Load switches and shorting bars are to be located within the confines of their respective fields. The combined total of switches and shorting bars shall be limited to a minimum of none and a maximum of four.

FIGURE 2. Panel markings.

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NOTES: 1. Component locations are relative, not dimensional.

2. Load switches and shorting bars are to be located in the NOTE 2 general area above.

FIGURE 3. Reference - desired layout.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-10308F	Test Set, Low Voltage Circuit, Automotive, 12 and 24 Volt DC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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